报告编号
Report No. LTI 504040 -V0

测 试 报 告

Test report

试验项目 Project	GAC T79-G2 HE Engine thermal shock test
客户零件号 Customer no.	/
委托单位 Requestor	佛吉亚排气控制技术开发（上海）有限公司研发中心 FCM-R&D Center
发布时间 Issued date	2026/8/21

版本 Version	修改内容及目的 Aim of the modification	日期 Date
0	创建报告 Report creating	2026/8/18

佛吉亚排气控制技术开发（上海）有限公司测试中心
FCM Test Center
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2.The report is only valid for tested samples.

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项目描述 Project description	GAC T79-G2 HE Engine thermal shock test		
测试方法 Test method	CHD-I-DSE-2007-Thermal Shock Endurance(1000cycles)		
项目名称 Project code	E10881D	项目阶段 Project phase	DV
客户 Customer	GAC	试验申请人 Requester (PDL)	QIAN Chris
测试位置 Test room number	发动机耐久台架 2 Durability Engine Bench 2	试验项目 Test item	振动耐久试验 Vibration Durability
发动机号 Engine no.	2-5#	燃油类型 Fuel type	汽油 Gasoline
测试环境温度（℃） Room temperature	26.9	相对湿度(%) Relative humidity	/
送样日期 Sample arrival date	2026/7/27	测试开始日期 Test start date	2026/7/31
试验周期 Test period	2026/07/31~8/16	试验通过否 Test passed	通过 PASS
结果 Results	1.□试验按照转速5500rpm，100%油门做为高温测点 Launched test at 5500rpm, 100% throttle as high temp condition 2.□测试了1000个循环后，试验样件无泄漏，无开裂 After 1000 cycles, part no leakage, no crack.		

作者Author		审核Check		批准Approval	
工程师Engineer's name	赵伟 ZHAO Wei	专家Expert's name	钱星辉 QIAN Chris	主管Leader's name	方云 FANG Yun
日期Date	2026/8/18	日期Date	2026/8/21	日期Date	2026/8/21

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附录 Appendix:

1. 试验方法 Test method

2. 设备信息 Equipment information

3. 样件信息 Sample information

4. 试验布置 Test setup

5. 试验数据 Test data

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1. 试验方法 Test method	
试验方法 Test method	CHD-I-DSE-2007-Thermal Shock Endurance(1000cycles)
试验流程描述 Test procedure description	振动耐久 Vibration durability: 正式试验前需要进行振动扫频以确认共振点信息 Previously testing vibration level for each vibration point to identify the main resonance. 根据申请者或者专家的结论进行一定时间的耐久试验。 Planning a durability vibration test from the previously reported resonance for a number of hours determined by the requester/durability expert.
目标 Target	目标 Target: 1. 样件无贯穿裂纹。 1. No through crack was found on sample. 2. 样件无泄漏。 2. No leakage was detected on sample. 3.载体无碎裂，无吹蚀。 3. No crack or erosion on canning.

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2. 设备信息 Equipment information

耐久台架 Durability bench	2#	设备编号 Gage number	L_TE_004
供应商 Supplier	D2t	测功机类型 Dyno property	电力测功机 AC dyno
最大功率 Power(kW)	300	最大扭矩 Torque(N·m)	480
最高转速 Speed(RPM)	10000	校准有效期 Calibration due date	2027/2/11



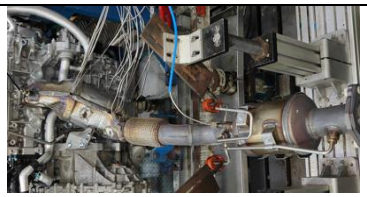
设备示意图（样件为举例，非本试验样件）
Equipment picture(sample is for illustration, not for the test)

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3. 样件信息 Sample information

样品名称 Sample name	样品编号 Sample no.	数量 Quantity	样品是否考核? Evaluate or not?	照片 Picture
Hot End	C004J-1.5T-GPF-9001-DV-002	1	考核 Evaluate	
Cold End	E11995B-1.5L-CE-9001-DV-001	1	不考核 Not	
Cold End	E11995B-1.5L-CE-9001-DV-002	1	不考核 Not	

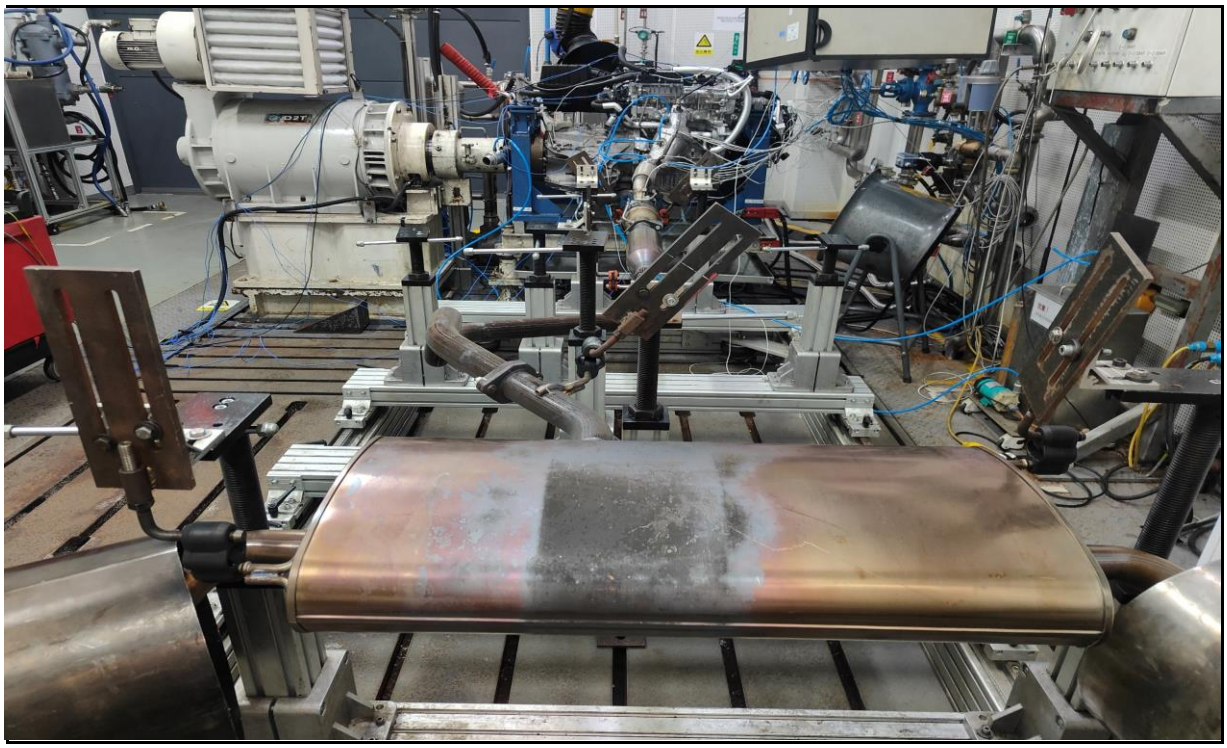
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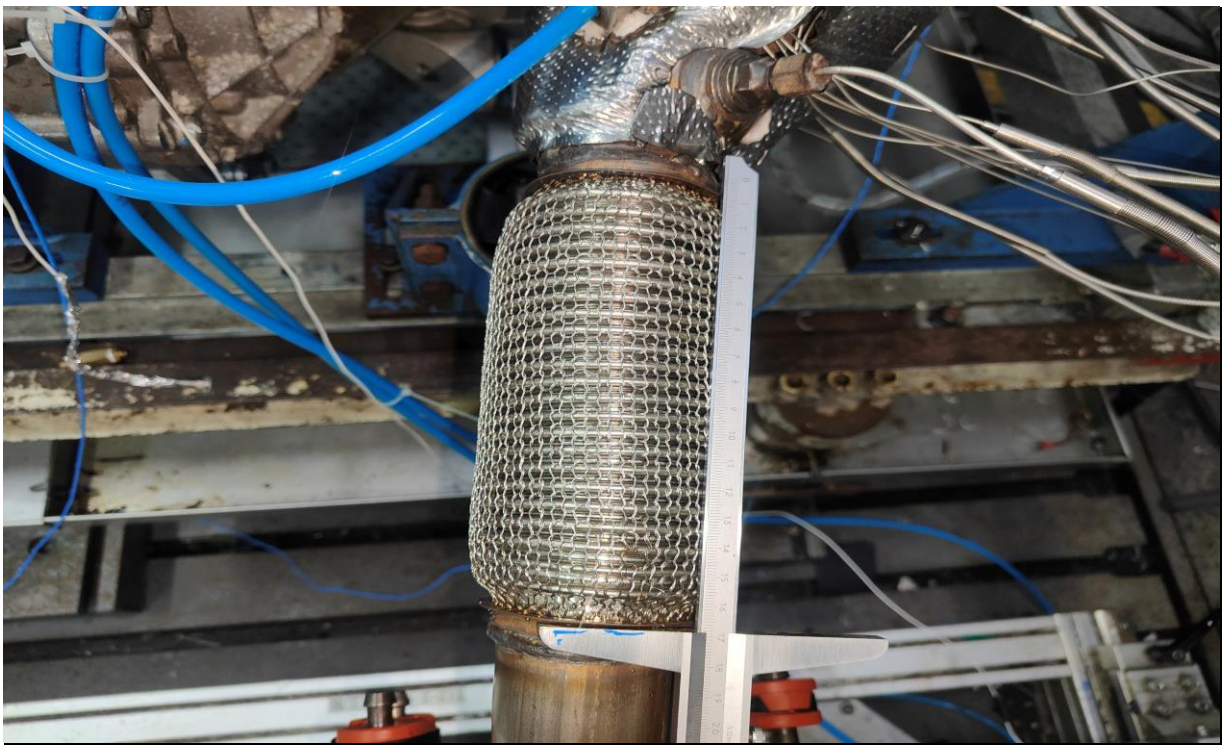
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4. 试验布置 Test setup

台架安装情况总览 Bench installing status overview

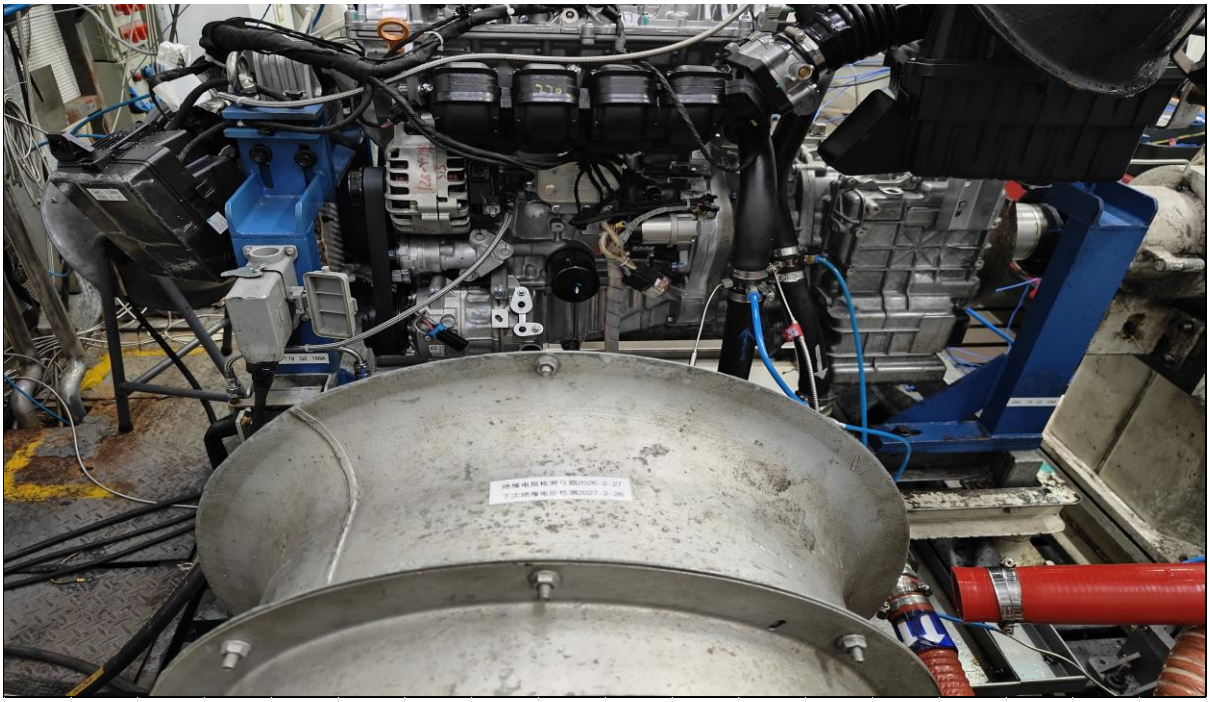




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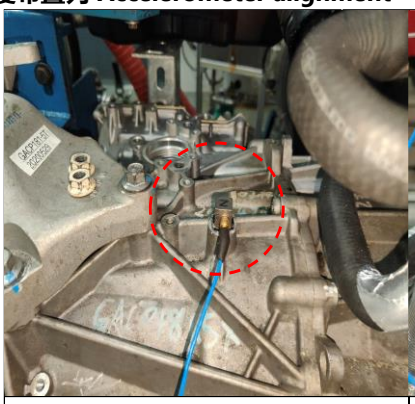
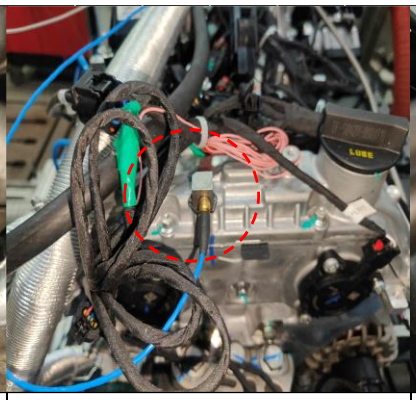
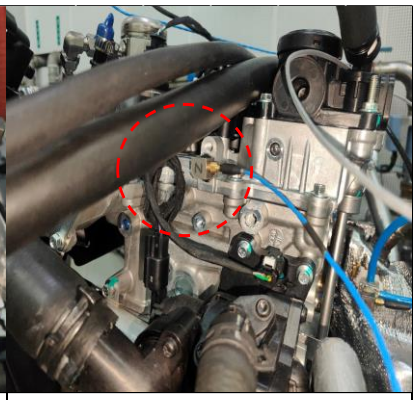
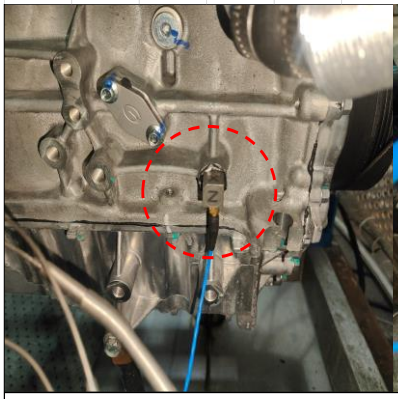
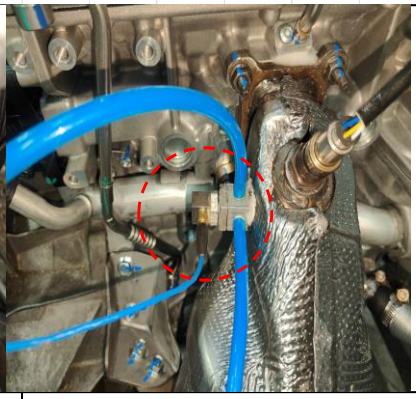
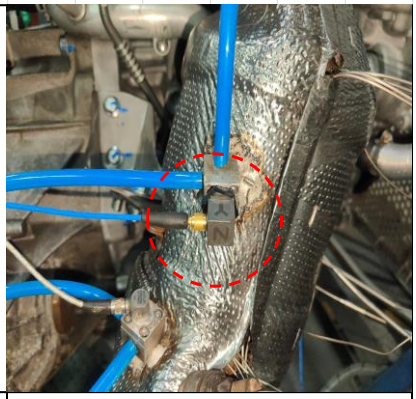
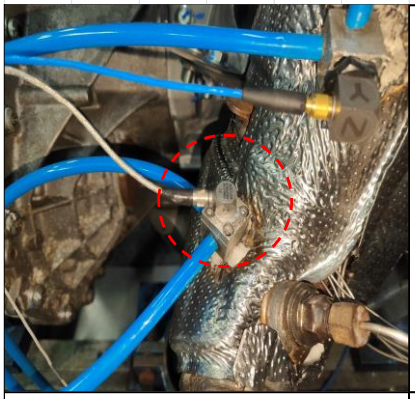


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加速度布置列 Accelerometer alignment

		
ENG101 变速箱	ENG102 缸盖（皮带轮侧）	ENG103 缸盖（变速箱侧）
		
ENG104 缸体	AC101 Inlet cone	AC102 TWC
		
AC103 Outlet cone		

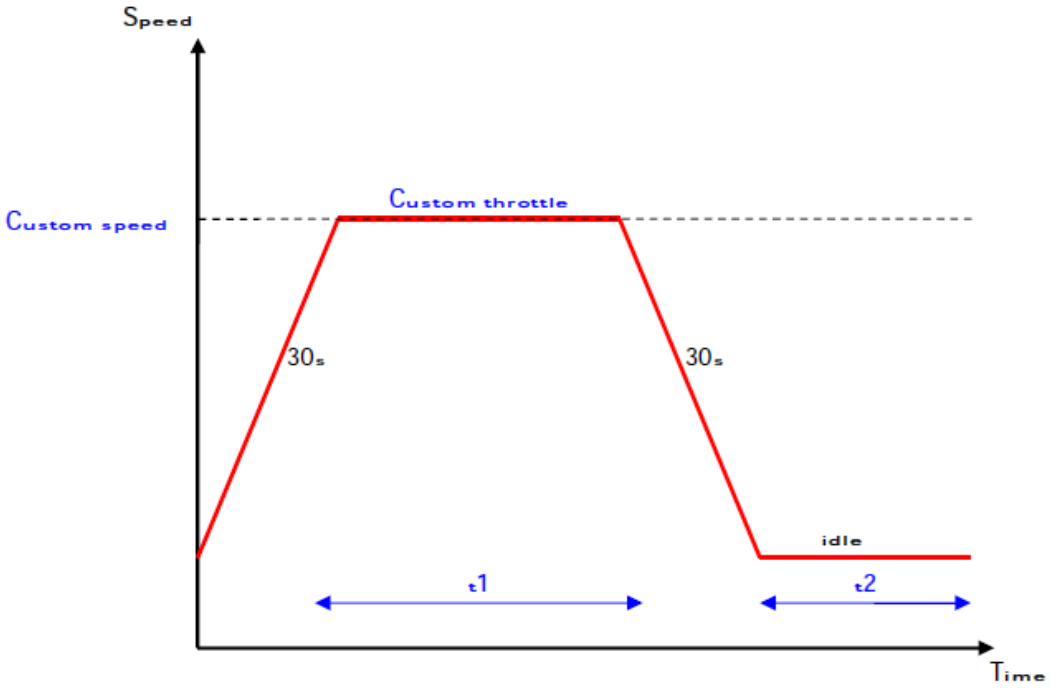
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试验工况 Test condition

热冲击耐久程序说明 Thermal shock test procedure description



佛吉亚标准热冲击试验流程@顺序5-9
Faurecia standard thermal shock procedure @ sequence 5-9

Parametrer	Value	Unit
Custom speed	5500	RPM
Custom throttle	100	%
T1	5	Min
T2	9	Min
Total test cycles	1000	Cycles

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5. 试验数据 Test data

样件检查结果 Sample checking result

观察 Observation

样件在0.3bar的压力下没有发现泄漏，满足≤0.5L/min的需求。
无损着色探伤未发现缺陷。

结论 Conclusion

样件在后处理分析阶段未发现缺陷。

泄露测试 Leakage test
耐久前 Before durability test





2. 总览

C004J-1.5T-GPF-9001-DV-002

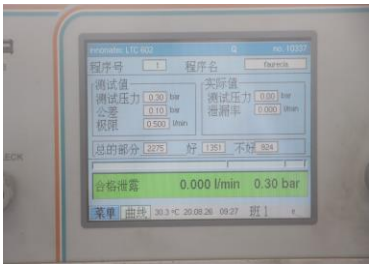




› 样件在收货状态下的全局视图。

3. 气密性检查









› 样件在0.3bar的压力下没有发现泄漏，满足≤0.5L/min的需求。

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4. 无损着色探伤



› 未发现缺陷。

4. 无损着色探伤



› 未发现缺陷。

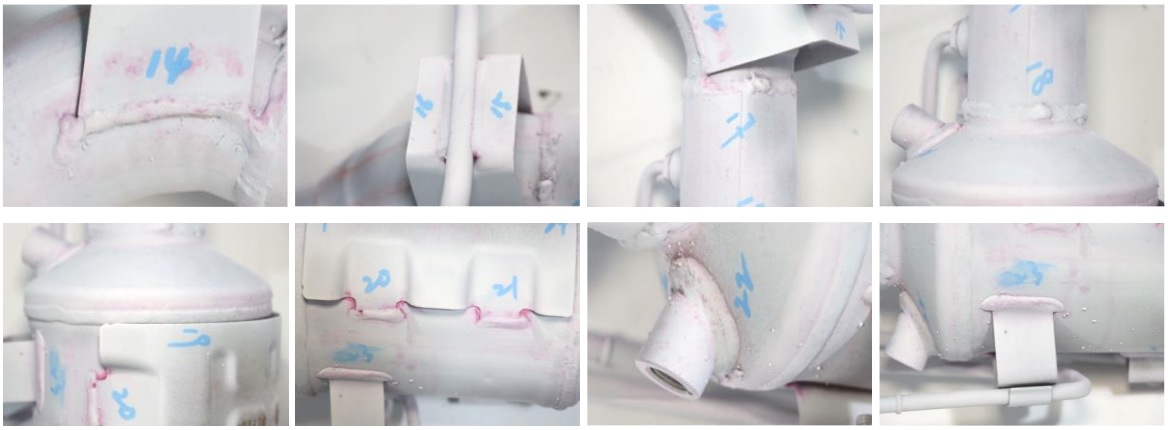
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4. 无损着色探伤

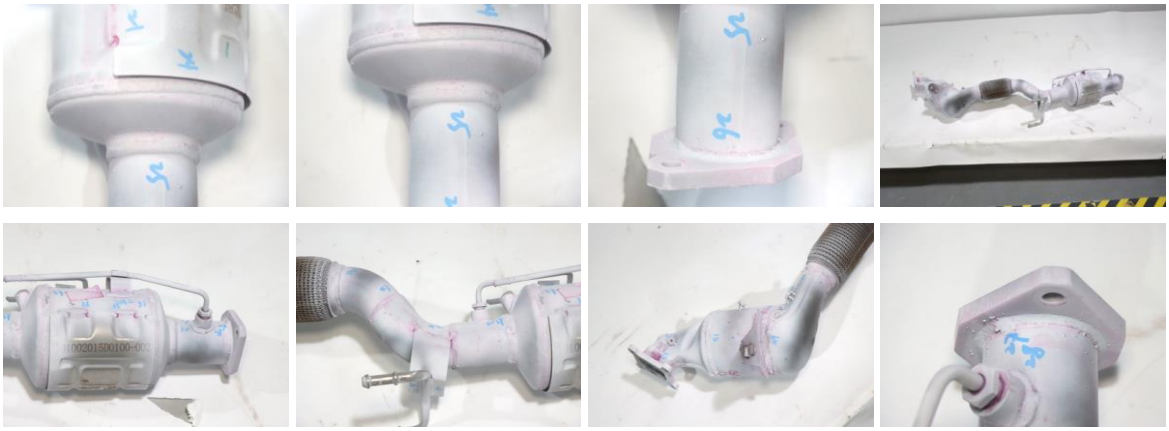
Internal & Partners



› 未发现缺陷。

4. 无损着色探伤

Internal & Partners



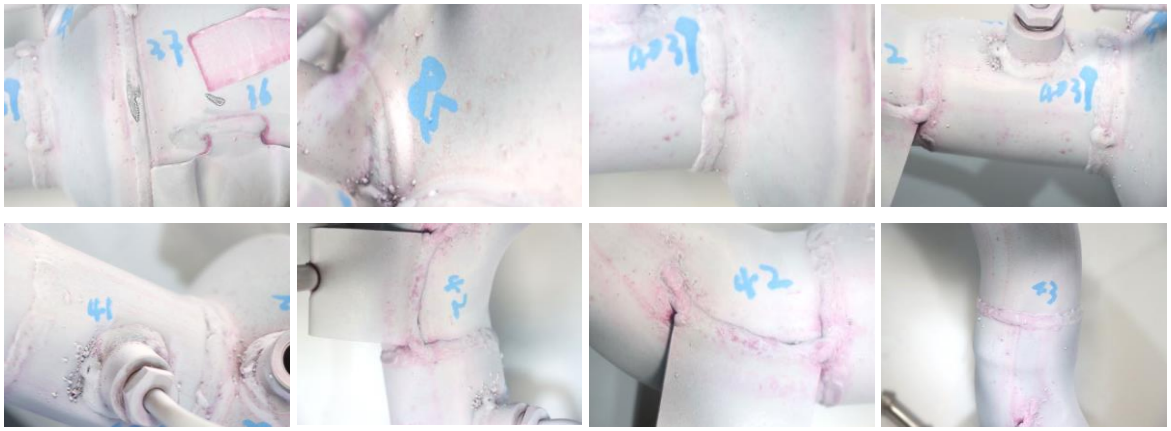
› 未发现缺陷。

4. 无损着色探伤



› 未发现缺陷。

4. 无损着色探伤



› 未发现缺陷。

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4. 无损着色探伤



› 未发现缺陷。

4. 无损着色探伤



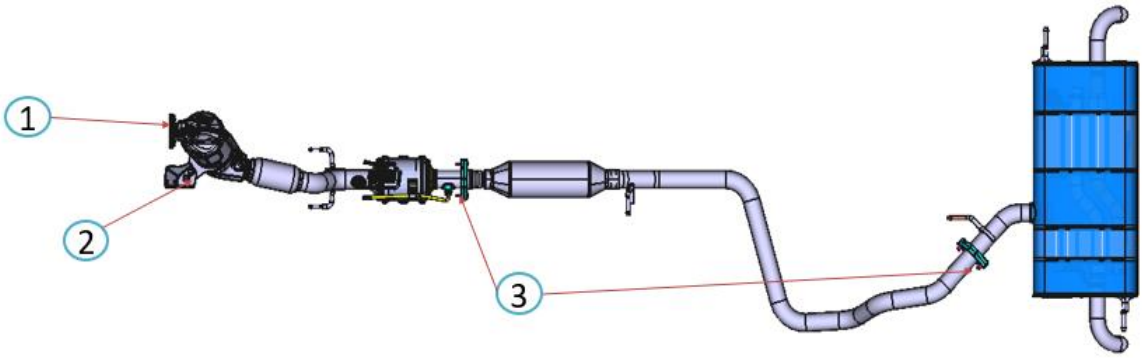
› 未发现缺陷。

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力矩检查结果 Torque checking result.



试验工况 Test condition			试验前 Before test	试验后 After test
安装力矩 Tightening torque (N·m)	1	25±2	25.1/26.3/25.3/25.5	20.1/19.2/25.3/31.2
	2	50±5	52.5/52.7/51.3/50.5	70.2/69.0/71.7/60.4
	3	50±5	52.2/52.3/52.3/54.2	70.9/50.5/69.8/55.6
	4	NA		
	5	NA		
	6	NA		
	7	NA		

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Tracking sheet试验跟踪记录表

试验日记 Log of the test

热冲击耐久跟踪表 Thermal shock endurance tracking sheet

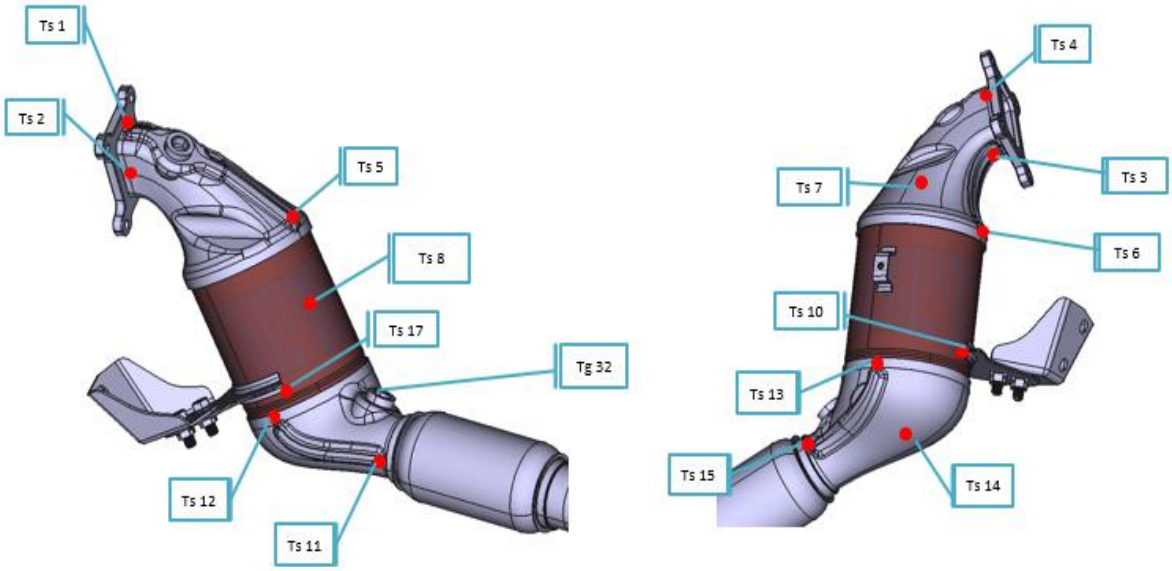
日期 Date	试验运行循环数 Procedure running cycles	发动机运行时间 Engine running time (h)	检查/事件/行动方案 Observations / Incidents / Actions
2026/7/31	0	27.9	排气系统安装 Exhaust system setup
2026/7/31	0	30.9	振动和温度扫频 Vibration and temperature measurement
2026/8/2	0	30.9	热冲击耐久10时11分:热冲击耐久转速为5500RPM,程序为5-9。 Thermal durability test 10h 11: finalize thermal vibration procedure at 5500 RPM, test sequence:5-9.
2026/8/12	703	222.9	目视检查及发动机维护保养 Visual inspection & engine maintenance
2026/8/16	1000	308.4	6:03 ： 热冲击耐久试验结束 6:03： thermal test finish

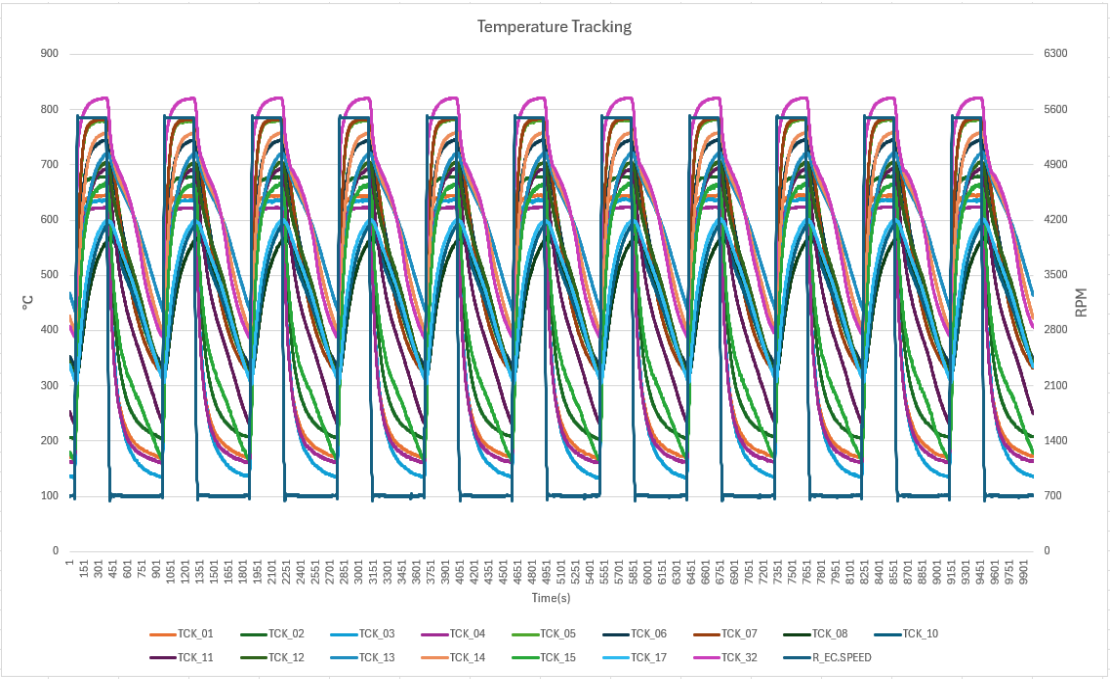
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热冲击耐久温度跟踪示意图 Critical points tracking sheet





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